

Name: Cyrish D. Lerio

Linux Commands Familiarization (Laboratory Exercise 2)

This laboratory exercise demonstrates how I performed Linux file commands to gain familiarity. This laboratory/lesson is provided by our teacher.

Objectives:

- Learn to navigate the Linux file system using basic commands.
- Gain experience creating, manipulating, and viewing files.
- Understand fundamental file and directory permissions.

Prerequisites:

Basic understanding of computers and operating systems.
Access to a Linux system (e.g., physical machine, virtual machine, online terminal)

Materials:

Terminal window
Text editor

Lab Instructions:

1. Logging In and Basic Navigation:

- Open a terminal window on your Linux system.
- Type `whoami` and press `Enter`. This command displays your username.
 - Type `pwd` and press `Enter`. This command shows your current working directory, which is your starting location in the file system.
 - Use the `cd` command to change directories. For example, type `cd Desktop` and press `Enter` to navigate to the "Desktop" directory.
 - Type `pwd` again to confirm your current directory has changed.
 - Use the `ls` command to list the contents of the current directory.
 - Use the `ls -l` command to view a detailed listing with additional information like file permissions and ownership.

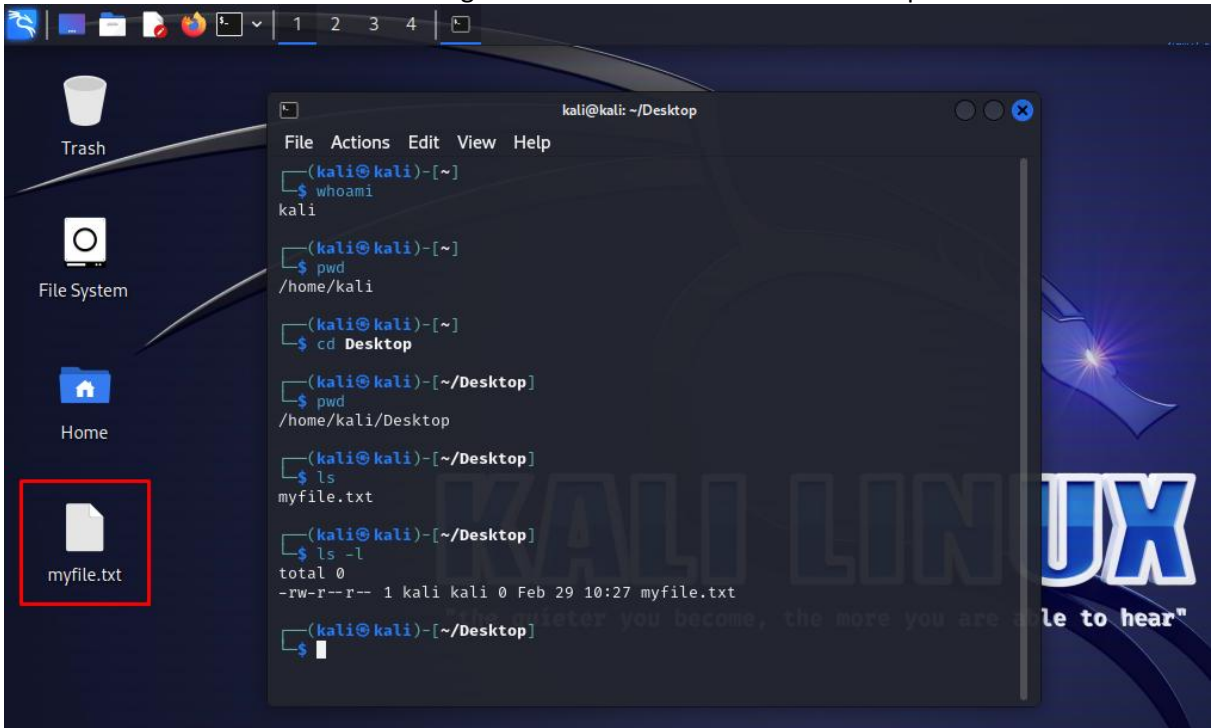


Figure 1. Kali Linux Terminal

2. Creating and Manipulating Files:

- Use the `touch` command to create a new file. For example, type `touch myfile.txt` and press `Enter` to create a new file named "myfile.txt".
- Use the `ls` command to verify that the file has been created.
- Use the `cat` command to view the contents of a file. For example, type `cat myfile.txt` and press `Enter`. (An empty file will display no content.)
- Use a text editor like nano or vi to edit the file.
- To open the file in nano, type `nano myfile.txt`.
- Once in the editor, type your desired content and save the changes by pressing `Ctrl+O` and then `Enter`.

- Exit the editor by pressing **Ctrl+X**.
- Use the **cat** command again to verify the changes made to the file.
- Use the **mv** command to rename a file. For example, type **mv myfile.txt renamed_file.txt** to rename the file to "renamed_file.txt".
- Use the **ls** command to confirm the file has been renamed.
- Use the **rm** command to delete a file with caution. For example, type **rm renamed_file.txt** to delete the renamed file. Be sure you no longer need the file before deleting it.

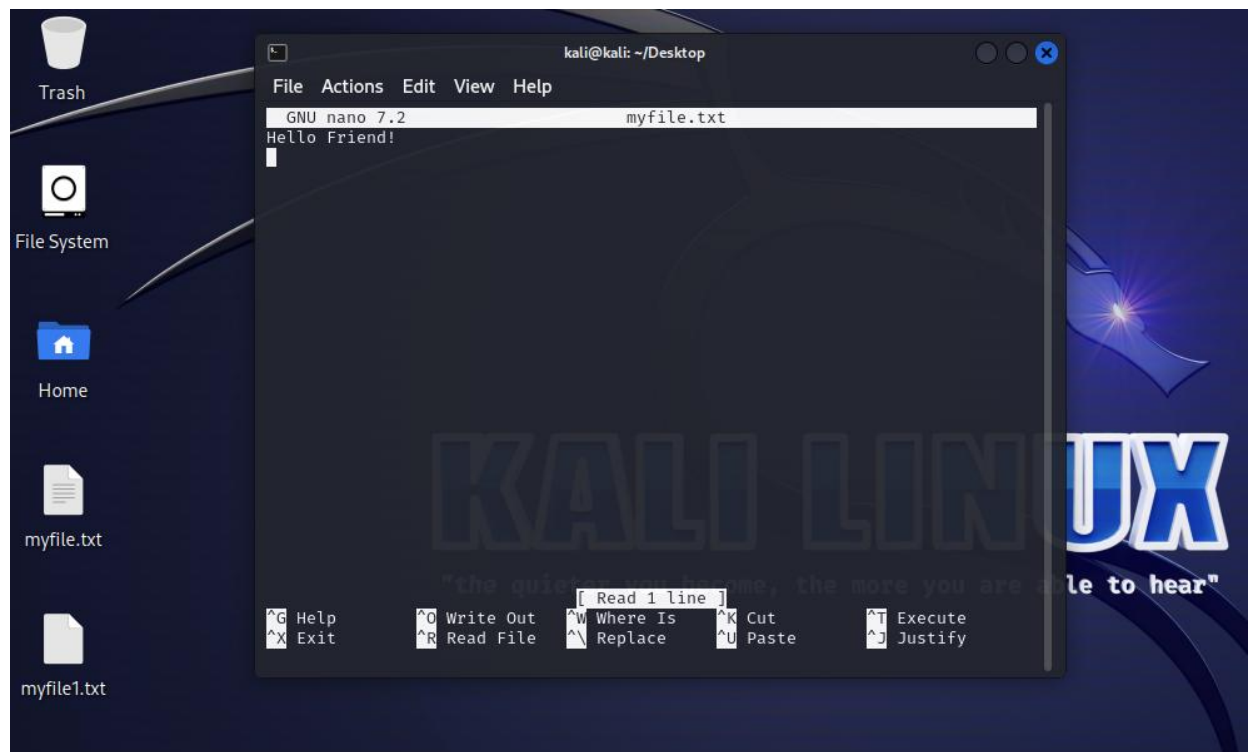


Figure 2. Text editor Nano

3. Understanding File Permissions:

- Use the **ls -l** command again to view detailed information about the files in your current directory.
- Pay attention to the first three characters at the beginning of each line, which represent the file permissions.
The first character represents the file type (e.g., - for regular file, d for directory).
The next three characters represent the permissions for three user groups:
 - **Owner (user):** **r** (read), **w** (write), **x** (execute)
 - **Group:** **r** (read), **w** (write), **x** (execute)
 - **Others:** **r** (read), **w** (write), **x** (execute)

Experiment with the **chmod** command to change file permissions with caution. However, we strongly recommend against modifying permissions during this beginner exercise. Modifying permissions incorrectly can have unintended consequences, so it's best to leave this for later exploration under proper guidance.

4. Cleaning Up:

- It's good practice to clean up after your exercises.
- Use the **rm** command to delete any unnecessary files you created during the exercise (e.g., **rm myfile.txt**).

Note:
Use the **man** command followed by a command name (e.g., **man cd**) to access the manual page for detailed information about the specific command.
Many online resources like tutorials and cheat sheets can be helpful for further learning.
Don't hesitate to experiment and explore, but be cautious and back up important data before making significant changes.

Challenge Exercise: Organize Your Music Collection

Scenario:

You have a downloaded music collection in a folder named "Music" on your Linux system. The music files are a mix of different formats (e.g., mp3, wav) and are not organized by artist or album.

Task:

Navigate to the "Music" directory and create at least 10 subdirectories for music genre (e.g. rock metal, alternative rock, ballad, k-pop, electronic pop, etc...) Inside each of the "Music" genre subdirectories create at least 3 subdirectories for artists belonging in the music genre. Inside each artist directory create at least 1 text file of their songs. For each text file write the lyrics of the song. Swap the contents of each "artist" subdirectories.Take screenshots for each task performed, paste it in this document and label appropriately. Have your work check by your instructor.

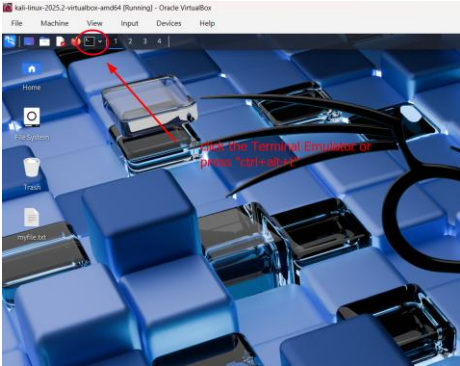
Remember:

Be cautious when using the mv command, as accidentally moving files to the wrong location can lead to difficulty finding them later.

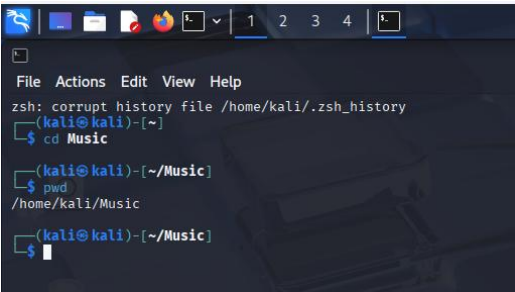
If you make a mistake, you can use the mv command again to move the file back to its original location.

You have completed the beginner-level exercises for Linux commands familiarization. Remember, this is just the beginning of your journey. Continue practicing and exploring to enhance your Linux skills.

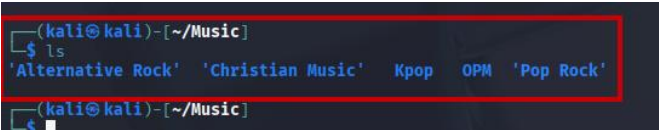
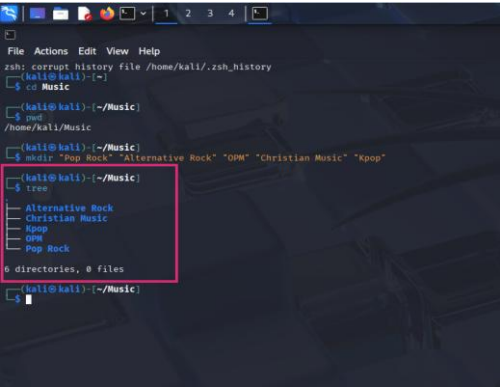
1. Click the Terminal Emulator.



2. I navigate into "Music" Directory.



3. I created only 5 sub-directories for Music Genre (at least 10).



4. Inside each of the “Music” genre subdirectories I create at least 3 subdirectories for artists belonging in the music genre.

```
File Actions Edit View Help
└─$ cd "Christian Music"
(kaliⓈkali)-[~/Music/Christian Music]
└─$ mkdir "Hillsong Worship" "Praise and Worship" "Don Moen"
(kaliⓈkali)-[~/Music/Christian Music]
└─$ cd
(kaliⓈkali)-[~]
└─$ cd Music
(kaliⓈkali)-[~/Music]
└─$ cd "Kpop"
(kaliⓈkali)-[~/Music/Kpop]
└─$ mkdir "Blackpink" "BTS" "Twice"
(kaliⓈkali)-[~/Music/Kpop]
└─$ cd
(kaliⓈkali)-[~]
└─$ cd Music
(kaliⓈkali)-[~/Music]
└─$ tree
.
├── Alternative Rock
│   ├── Arctic Monkeys
│   ├── Wave To Earth
│   └── Yung Kai
├── Christian Music
│   ├── Don Moen
│   ├── Hillsong Worship
│   └── Praise and Worship
├── Kpop
│   ├── Blackpink
│   ├── BTS
│   └── Twice
├── OPM
│   ├── Ben8Ben
│   ├── Cup of Joe
│   └── December Avenue
└── Pop Rock
    ├── Olivia Rodrigo
    ├── Sabrina Carpenter
    └── Taylor Swift

21 directories, 0 files
(kaliⓈkali)-[~/Music]
└─$
```

5. Inside each artist directory I create at least 1 text file of their songs.

```
(kali@kali) [~]
$ cd Music

(kali@kali)-[~/Music]
$ cd "OPM/Cup of JOe"
cd: no such file or directory: OPM/Cup of JOe

(kali@kali)-[~/Music]
$ cd "OPM/Cup of Joe"

(kali@kali)-[~/Music/OPM/Cup of Joe]
$ cat tingin.txt
Cup of Joe - Tingin (Lyrics)

(kali@kali)-[~/Music/OPM/Cup of Joe]
$ nano tingin.txt

(kali@kali)-[~/Music/OPM/Cup of Joe]
```

```
(kali@kali)-[~/Music]
$ tree
.
├── Alternative Rock
│   ├── Arctic Monkeys
│   │   └── yours.txt
│   ├── Wave To Earth
│   │   └── love.txt
│   └── Yung Kai
│       └── blue.txt
├── Christian Music
│   ├── Don Moen
│   │   └── makeaway.txt
│   ├── Hillsong Worship
│   │   └── still.txt
│   └── Praise and Worship
│       └── reasons.txt
├── Kpop
│   ├── Blackpink
│   │   └── kill_this_love.txt
│   ├── BTS
│   │   └── butter.txt
│   └── Twice
│       └── tt.txt
├── OPM
│   ├── Ben&Ben
│   │   └── make.txt
│   ├── Cup of Joe
│   │   └── tingin.txt
│   └── December Avenue
│       └── eroplanong_papel.txt
└── Pop Rock
    ├── Olivia Rodrigo
    │   └── vampire.txt
    ├── Sabrina Carpenter
    │   └── please.txt
    └── Taylor Swift
        └── enchanted.txt

21 directories, 15 files

(kali@kali)-[~/Music]
$
```


6. For each text file I write the lyrics of the song.

```
File Actions Edit View Help
└─$ cd
(kali@kali)-[~]
└─$ cd Music
(kali@kali)-[~/Music]
└─$ cd "OPM/Cup of Joe"
cd: no such file or directory: OPM/Cup of JOe
(kali@kali)-[~/Music]
└─$ cd "OPM/Cup of Joe"
(kali@kali)-[~/Music/OPM/Cup of Joe]
└─$ cat tingin.txt
Cup of Joe - Tingin (Lyrics)
(kali@kali)-[~/Music/OPM/Cup of Joe]
└─$ nano tingin.txt
(kali@kali)-[~/Music/OPM/Cup of Joe]
└─$ txt
Command 'txt' not found, did you mean:
  command 'ttx' from deb fonttools
  command 'tgt' from deb tcm
  command 'tt' from deb treetop
Try: sudo apt install <deb name>
(kali@kali)-[~/Music/OPM/Cup of Joe]
└─$ cat tingin.txt
Cup of Joe - Tingin (Lyrics)

Dahan-dahan lang sa gitna man ng daan
Mga saglit na inilikha
Kakaiba ang tama
Ng sinag sa 'yong kutis na kayumanggi
Oh, sa'n ba 'ko dinadala?
Bawat ngiting biglaang nabura
Iyong naipinta
Hiwaga ng 'yong tinging nang-aalipin
Kahit sa'n man madala
'Di pinapansin ingay sa tabi
Magulong kapaligiran, sa 'yo lang ang tingin
'Di pinapansin, ika'y paiikutin
Nang dahan-dahan lang sa gitna man ng daan
Sa bawat sandaling ikaw ay pinagmamasdan
May dumadapong kiliti na 'di maunawaan
Walang imik, 'di mabanggit na sa aking isip
Ikaw lang ang nagmamarka
Kahit mabitin aking salita
Mata'y ibinubunyag na
```

```
kali@kali: ~/
File Actions Edit View Help

21 directories, 0 files
(kali@kali)-[~/Music]
└─$ cd
(kali@kali)-[~]
└─$ cd Music
(kali@kali)-[~/Music]
└─$ cd "Pop Rock/Olivia Rodrigo"
(kali@kali)-[~/Music/Pop Rock/Olivia Rodrigo]
└─$ nano vampire.txt
(kali@kali)-[~/Music/Pop Rock/Olivia Rodrigo]
└─$ cat vampire.txt
Olivia Rodrigo - Vampire (Lyrics)

I hate to give the satisfaction askin' how you're doin' now
How's the castle built off people you pretend to care about?
Just what you wanted
Look at you, cool guy, you got it
I see the parties and the diamonds sometimes when I close my eyes
Six months of torture you sold as some forbidden paradise
I loved you truly
You gotta laugh at the stupidity
'Cause I've made some real big mistakes
But you make the worst one look fine
I should've known it was strange
You only come out at night
I used to think I was smart
But you made me look so naive
The way you sold me for parts
As you sunk your teeth into me, oh
Bloodsucker, fame-fucker
Bleedin' me dry like a goddamn vampire
And every girl I ever talked to told me you were bad, bad news
You called them crazy, god, I hate the way I called them crazy too
You're so convincin', how do you lie without flinchin'?
(How do you lie? How do you lie? How do you lie?)
Ooh, what a mesmerizin', paralyzin', fucked-up little thrill
Can't figure out just how you do it, and God knows I never will
Went for me, and not her
'Cause girls your age know better
I've made some real big mistakes
But you make the worst one look fine
I should've known it was strange
You only come out at night
I used to think I was smart
But you made me look so naive
```

7. I swap the contents of each “artist” subdirectories.

```
(kali@kali)-[~/Music]
$ tree
.
├── Alternative Rock
│   ├── Arctic Monkeys
│   │   └── yours.txt
│   ├── Wave To Earth
│   │   └── love.txt
│   └── Yung Kai
│       └── blue.txt
├── Christian Music
│   ├── Don Moen
│   │   └── makeaway.txt
│   ├── Hillsong Worship
│   │   └── still.txt
│   └── Praise and Worship
│       └── reasons.txt
├── Kpop
│   ├── Blackpink
│   │   └── kill_this_love.txt
│   ├── BTS
│   │   └── butter.txt
│   └── Twice
│       └── tt.txt
├── OPM
│   ├── Ben&Ben
│   │   └── make.txt
│   ├── Cup of Joe
│   │   └── tingin.txt
│   ├── December Avenue
│   │   └── eroplanong_papel.txt
│   └── Pop Rock
│       ├── Olivia Rodrigo
│       │   └── vampire.txt
│       ├── Sabrina Carpenter
│       │   └── please.txt
│       └── Taylor Swift
│           └── enchanted.txt
└── 21 directories, 15 files

(kali@kali)-[~/Music]
```

swap the contents of each "artist" subdirectories.

```
(kali@kali)-[~/Music]
$ cd Music

(kali@kali)-[~/Music]
$ pwd
/home/kali/Music

(kali@kali)-[~/Music]
$ ls "Alternative Rock"
Arctic Monkeys  Wave To Earth  Yung Kai

(kali@kali)-[~/Music]
$ cd "Alternative Rock"

(kali@kali)-[~/Music/Alternative Rock]
$ mkdir temp_swap

(kali@kali)-[~/Music/Alternative Rock]
$ mv "Arctic Monkeys"/* temp_swap/

(kali@kali)-[~/Music/Alternative Rock]
$ mv "Yung Kai"/* "Arctic Monkeys"/

(kali@kali)-[~/Music/Alternative Rock]
$ mv temp_swap/* "Yung Kai"/

(kali@kali)-[~/Music/Alternative Rock]
$ rmdir temp_swap

(kali@kali)-[~/Music/Alternative Rock]
$ tree
.
├── Arctic Monkeys
│   ├── blue.txt
│   ├── Wave To Earth
│   │   └── love.txt
│   └── Yung Kai
│       └── yours.txt
└── 21 directories, 15 files

(kali@kali)-[~/Music]
```

```
File Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ cd Music

(kali@kali)-[~/Music]
$ pwd
/home/kali/Music

(kali@kali)-[~/Music]
$ cd ~/Music/"Pop Rock"

(kali@kali)-[~/Music/Pop Rock]
$ mv "Olivia Rodrigo/vampire.txt" "Sabrina Carpenter/"

(kali@kali)-[~/Music/Pop Rock]
$ mv "Sabrina Carpenter/please.txt" "Taylor Swift/"

(kali@kali)-[~/Music/Pop Rock]
$ mv "Taylor Swift/enchanted.txt" "Olivia Rodrigo/"

(kali@kali)-[~/Music/Pop Rock]
$ cd
```

```
File Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ cd Music

(kali@kali)-[~/Music]
$ pwd
/home/kali/Music

(kali@kali)-[~/Music]
$ tree
.
├── Alternative Rock
│   ├── Arctic Monkeys
│   │   ├── blue.txt
│   │   ├── Wave To Earth
│   │   │   └── love.txt
│   │   └── Yung Kai
│   │       └── yours.txt
│   ├── Christian Music
│   │   ├── Don Moen
│   │   │   └── still.txt
│   │   ├── Hillsong Worship
│   │   │   └── reasons.txt
│   │   └── Praise and Worship
│   │       └── makeaway.txt
│   ├── Kpop
│   │   ├── Blackpink
│   │   │   └── butter.txt
│   │   ├── BTS
│   │   │   └── tt.txt
│   │   └── Twice
│   │       └── kill_this_love.txt
│   ├── OPM
│   │   ├── Ben&Ben
│   │   │   └── tingin.txt
│   │   ├── Cup of Joe
│   │   │   └── eroplanong_papel.txt
│   │   ├── December Avenue
│   │   │   └── make.txt
│   │   └── Pop Rock
│   │       ├── Olivia Rodrigo
│   │       │   ├── enchanted.txt
│   │       │   ├── Sabrina Carpenter
│   │       │   │   └── vampire.txt
│   │       │   └── Taylor Swift
│   │       │       └── please.txt
│   └── 21 directories, 15 files

(kali@kali)-[~/Music]
```

after swap